Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2013 – Paper 3 – Solutions**

June 2013 – Question 1 – Solution

(a) Let $\boldsymbol{p}$ and $\boldsymbol{q}$ be two propositions.

(i) State the converse of $(\boldsymbol{p} \wedge \boldsymbol{q}) \rightarrow (\boldsymbol{q} \vee \sim\boldsymbol{p})$. [1]

Statement: $(\boldsymbol{p} \wedge \boldsymbol{q}) \rightarrow (\boldsymbol{q} \vee \sim\boldsymbol{p})$

Converse: $(\boldsymbol{q} \vee \sim\boldsymbol{p}) \rightarrow (\boldsymbol{p} \wedge \boldsymbol{q})$

(ii) Show that the contrapositive of the inverse of $(\boldsymbol{p} \wedge \boldsymbol{q}) \rightarrow (\boldsymbol{q} \vee \sim\boldsymbol{p})$ is the converse of $(\boldsymbol{p} \wedge \boldsymbol{q}) \rightarrow (\boldsymbol{q} \vee \sim\boldsymbol{p})$. [3]

Statement: $(\boldsymbol{p} \wedge \boldsymbol{q}) \rightarrow (\boldsymbol{q} \vee \sim\boldsymbol{p})$

Inverse: $\sim(\boldsymbol{p} \wedge \boldsymbol{q}) \rightarrow \sim(\boldsymbol{q} \vee \sim\boldsymbol{p})$

Contrapositive of the inverse: $(\boldsymbol{q} \vee \sim\boldsymbol{p}) \rightarrow (\boldsymbol{p} \wedge \boldsymbol{q})$ $\rightarrow$ Expression 1

Statement: $(\boldsymbol{p} \wedge \boldsymbol{q}) \rightarrow (\boldsymbol{q} \vee \sim\boldsymbol{p})$

Converse: $(\boldsymbol{q} \vee \sim\boldsymbol{p}) \rightarrow (\boldsymbol{p} \wedge \boldsymbol{q})$ $\rightarrow$ Expression 2

Since Expression 1 and Expression 2 are equal to each other, the contrapositive of the inverse is the converse.

Q.E.D.

(b) Solve the equation $\log_2(x+3) = 3 - \log_2(x+2)$. [5]

$$\log_2(x+3) = 3 - \log_2(x+2)$$

$$\log_2(x+3) + \log_2(x+2) = 3$$

$$\log_2(x+3)(x+2) = 3$$

Converting to exponential form gives:

$$(x+3)(x+2) = 2^3$$

$$x^2 + 2x + 3x + 6 = 8$$

$$x^2 + 5x + 6 - 8 = 0$$

$$x^2 + 5x - 2 = 0$$ which is in the form $ax^2 + bx + c = 0$,

where $a = 1, b = 5$ and $c = -2$.

Using the quadratic formula,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(5) \pm \sqrt{(5)^2 - 4(1)(-2)}}{2(1)}$$

$$= \frac{-5 \pm \sqrt{25+8}}{2}$$

$$= \frac{-5 \pm \sqrt{33}}{2}$$

The domain is $x > -2$.

Only $x = \frac{-5+\sqrt{33}}{2}$ satisfy $x > -2$, so it is valid.

$$\therefore x = \frac{-5+\sqrt{33}}{2}$$

(c) The amount of impurity, A, present in a chemical depends on the time it takes to purify. It is known that $A = 3e^{4t} - 7e^{2t} - 6$ at any time t minutes. Find the time at which the chemical is free of impurity (that is when $A = 0$). [6]

$A = 3e^{4t} - 7e^{2t} - 6$

When $A = 0$,

$3e^{4t} - 7e^{2t} - 6 = 0$

Let $y = e^{2t}$.

$3y^2 - 7y - 6 = 0$

$(3y + 2)(y - 3) = 0$

Either $3y + 2 = 0$ or $y - 3 = 0$

$3y = -2$ $\quad$ $y = 3$

$y = -\frac{2}{3}$

Consider:

$y = -\frac{2}{3}$

$e^{2t} = -\frac{2}{3}$

(No solutions)

Now,

Consider:

$$y = 3$$

$$e^{2t} = 3$$

$$\ln e^{2t} = \ln 3$$

$$2t \ln e = \ln 3$$

$$2t = \ln 3$$

$$t = \frac{\ln 3}{2}$$

$\therefore$ The time at which the chemical is free of impurity is $\frac{\ln 3}{2}$ minutes.

(d) On the same axes, sketch the graphs of $f(x) = 2x + 3$ and $g(x) = |2x + 3|$.

Show clearly ALL intercepts that may be present. [5]

$y = 2x + 3$

When $x = 0$,

$y = 2(0) + 3$

$y = 3$

The graph of $f(x)$ intersects the y-axis at point $(0, 3)$.

When $y = 0$,

$0 = 2x + 3$

$-3 = 2x$

$-\frac{3}{2} = x$

The graph of $f(x)$ intersects the x-axis at point $\left(-\frac{3}{2}, 0\right)$.

Now,

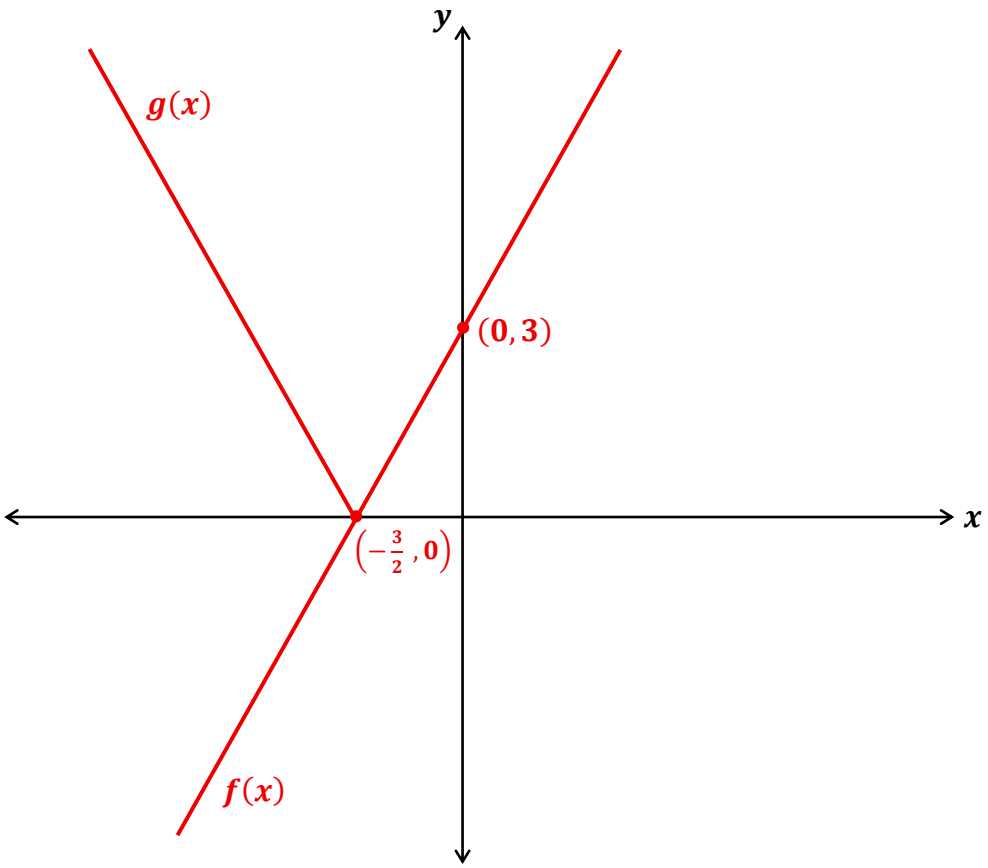
The graph of $g(x) = |2x + 3|$ is obtained by reflecting the part of $f(x) = 2x + 3$

where $x < -\frac{3}{2}$ and is below the x-axis, above the x-axis.

The graph of $g(x)$ intersects the y-axis at point $(0, 3)$.

The graph of $g(x)$ intersects the x-axis at point $\left(-\frac{3}{2}, 0\right)$.

The sketch below shows the graphs of $f(x) = 2x + 3$ and $g(x) = |2x + 3|$.



Total: 20 marks

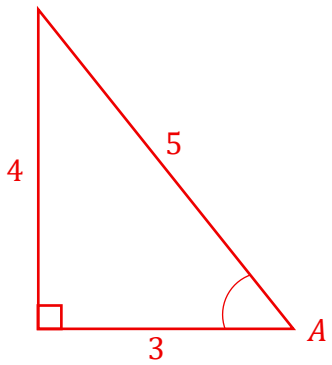
June 2013 – Question 2 – Solution

(a) A is an acute angle and B is an obtuse angle, where $\sin(A) = \frac{4}{5}$ and $\cos(B) = -\frac{3}{5}$.

Without finding the values of angles A and B, calculate $\cos(3A)$. [5]

We are given that $\sin(A) = \frac{4}{5}$.

Consider the diagram below.



By Pythagoras' Theorem,

$$a^2 + b^2 = c^2$$

$$a^2 + (4)^2 = (5)^2$$

$$a^2 + 16 = 25$$

$$a^2 = 25 - 16$$

$$a^2 = 9$$

$$a = \sqrt{9}$$

$$a = 3$$

Now,

$$\cos(A) = \frac{adj}{hyp}$$

$$= \frac{3}{5}$$

Using the addition formula,

$$\cos(3A) = \cos(2A + A)$$

$$= \cos 2A \cos A - \sin 2A \sin A$$

$$= (2\cos^2 A - 1)(\cos A) - (2\sin A \cos A)(\sin A)$$

$$= 2\cos^3 A - \cos A - 2\sin^2 A \cos A$$

$$= 2\left(\frac{3}{5}\right)^3 - \left(\frac{3}{5}\right) - 2\left(\frac{4}{5}\right)^2\left(\frac{3}{5}\right)$$

$$= 2\left(\frac{27}{125}\right) - \frac{3}{5} - 2\left(\frac{16}{25}\right)\left(\frac{3}{5}\right)$$

$$= \frac{54}{125} - \frac{3}{5} - \frac{96}{125}$$

$$= -\frac{117}{125}$$

$$\therefore \cos(3A) = -\frac{117}{125}$$

(b) Solve the equation $4\cos 2\theta - 14\sin\theta = 7$ for values of θ between 0 and 2π radians. [8]

$$4\cos 2\theta - 14\sin\theta = 7$$

$$4(1 - 2\sin^2\theta) - 14\sin\theta = 7$$

$$4 - 8\sin^2\theta - 14\sin\theta = 7$$

$$-8\sin^2\theta - 14\sin\theta + 4 - 7 = 0$$

$$-8\sin^2\theta - 14\sin\theta - 3 = 0$$

$(\times -1)$

$$8\sin^2\theta + 14\sin\theta + 3 = 0$$

$$(4\sin\theta + 1)(2\sin\theta + 3) = 0$$

Either $4\sin\theta + 1 = 0$ or $2\sin\theta + 3 = 0$

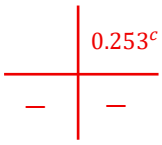
$4\sin\theta = -1$ $\quad$ $2\sin\theta = -3$

$\sin\theta = -\frac{1}{4}$ $\quad$ $\sin\theta = -\frac{3}{2}$

(no solutions since $-1 \le \sin\theta \le 1$)

Consider,

$\sin\theta = -\frac{1}{4}$

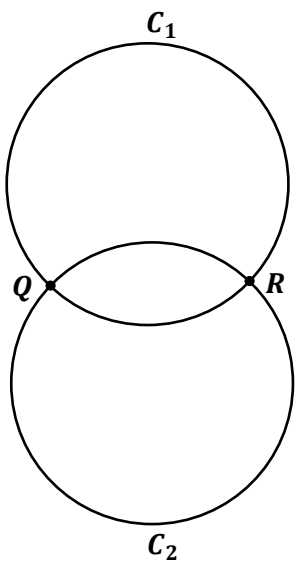


$$\theta = \pi + 0.253^c, 2\pi - 0.253^c$$

$$\theta = 3.39^c, 6.03^c$$

$\therefore \theta = 3.39^c, 6.03^c$ for $0 \le \theta \le 2\pi$

(c) An engineer is asked to build a table in the shape of two circles C_1 and C_2 which intersect each other, as shown in the diagram below (**not drawn to scale**).



The equations of C_1 and C_2 are $x^2 + y^2 + 4x + 6y - 3 = 0$ and $x^2 + y^2 + 4x + 2y - 7 = 0$ respectively.

A leg of the table is attached at EACH of the points Q and R where the circles intersect.

Determine the coordinates of the positions of the legs of the table. [7]

C_1: $x^2 + y^2 + 4x + 6y - 3 = 0$ → Equation 1

C_2: $x^2 + y^2 + 4x + 2y - 7 = 0$ → Equation 2

Equating both equations gives:

$$x^2 + y^2 + 4x + 6y - 3 = x^2 + y^2 + 4x + 2y - 7$$

$$6y - 3 = 2y - 7$$

$$6y - 2y = 3 - 7$$

$$4y = -4$$

$$y = \frac{-4}{4}$$

$$y = -1$$

Substituting $y = -1$ into Equation 1 gives:

$$x^2 + (-1)^2 + 4x + 6(-1) - 3 = 0$$

$$x^2 + 1 + 4x - 6 - 3 = 0$$

$$x^2 + 4x - 8 = 0$$ which is in the form $ax^2 + bx + c = 0$,

where $a = 1$, $b = 4$ and $c = -8$.

Using the quadratic formula,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(4) \pm \sqrt{(4)^2 - 4(1)(-8)}}{2(1)}$$

$$= \frac{-4 \pm \sqrt{16 + 32}}{2}$$

$$= \frac{-4 \pm \sqrt{48}}{2}$$

$$= \frac{-4 \pm \sqrt{16 \times 3}}{2}$$

$$= \frac{-4 \pm 4\sqrt{3}}{2}$$

$$= \frac{2(-2 \pm 2\sqrt{3})}{2}$$

$$= -2 \pm \sqrt{12}$$

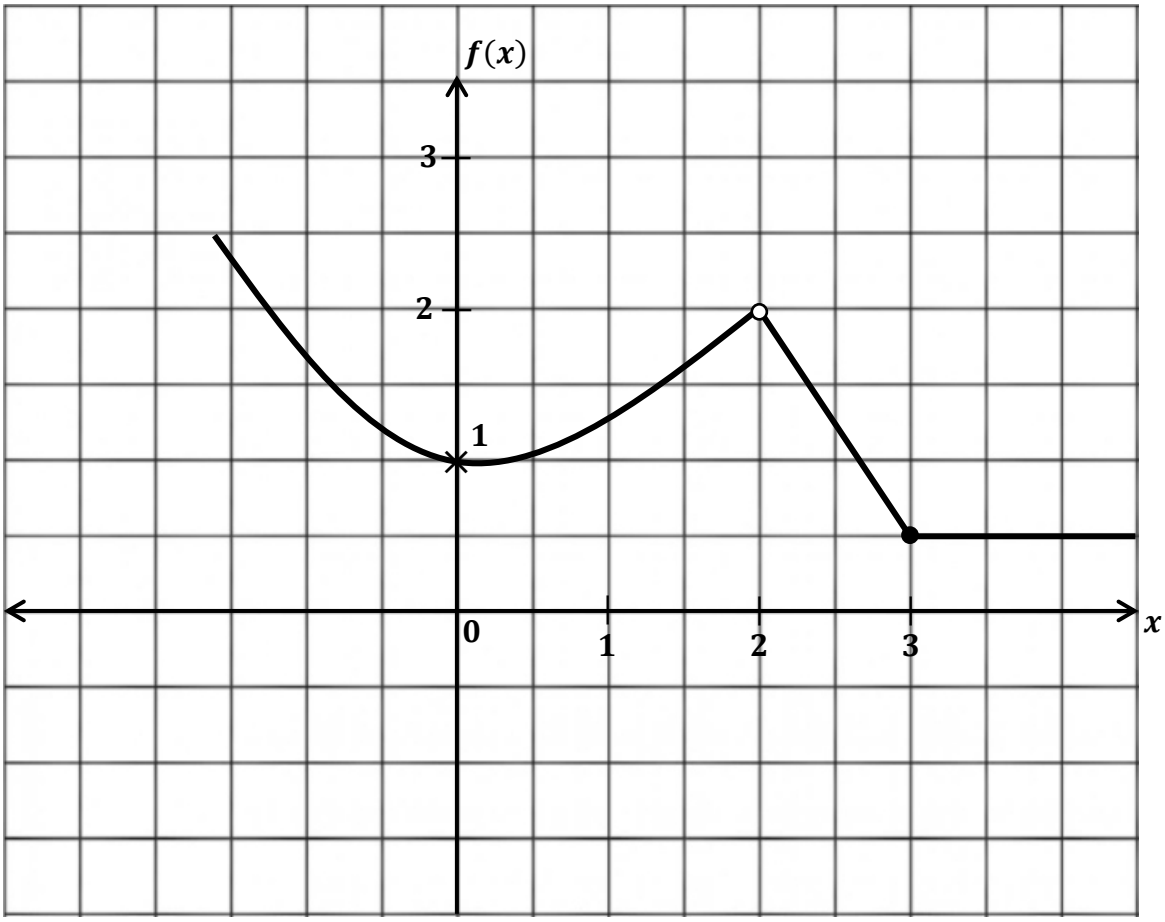
∴ The coordinates of the positions of the legs of the table are:

$Q = (-2 - 2\sqrt{3}, -1)$ and $R = (-2 + 2\sqrt{3}, -1)$

Total: 20 marks

June 2013 – Question 3 – Solution

(a) The diagram below shows the graph of a function, $f(x)$.



(i) Determine for the function

(a) $\lim_{x \to 0} f(x)$ [1]

From the graph, $\lim_{x \to 0} f(x) = 1$.

(b) $\lim_{x \to 2} f(x)$ [2]

From the graph, $\lim_{x \to 2} f(x) = 2$.

(ii) State whether f is continuous at $x = 2$. **Justify your answer.** [2]

There is an open circle at $(2, 2)$ which means that $f(2)$ is not defined.

In order for f to be continuous at $x = 2$, $f(2)$ must be defined.

Hence, f is not continuous at $x = 2$.

(b) Differentiate $f(x) = \frac{1}{\sqrt{2x}}$ from first principles. [5]

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

$$= \lim_{h\to 0} \frac{\frac{1}{\sqrt{2(x+h)}} - \frac{1}{\sqrt{2x}}}{h}$$

$$= \lim_{h\to 0} \frac{\frac{1}{\sqrt{2x+2h}} - \frac{1}{\sqrt{2x}}}{h}$$

$$= \lim_{h\to 0} \left[\left(\frac{1}{\sqrt{2x+2h}} - \frac{1}{\sqrt{2x}}\right) \div h\right]$$

$$= \lim_{h\to 0} \left[\left(\frac{\sqrt{2x}-\sqrt{2x+2h}}{\sqrt{2x}\sqrt{2x+2h}}\right) \times \frac{1}{h}\right]$$

$$= \lim_{h\to 0} \left[\left(\frac{\sqrt{2x}-\sqrt{2x+2h}}{\sqrt{2x}\sqrt{2x+2h}} \times \frac{\sqrt{2x}+\sqrt{2x+2h}}{\sqrt{2x}+\sqrt{2x+2h}}\right) \times \frac{1}{h}\right]$$

$$= \lim_{h\to 0} \left[\left(\frac{2x-(2x+2h)}{(\sqrt{2x}\sqrt{2x+2h})(\sqrt{2x})+(\sqrt{2x}\sqrt{2x+2h})(\sqrt{2x+2h})}\right) \times \frac{1}{h}\right]$$

$$= \lim_{h\to 0} \left[\left(\frac{2x-2x-2h}{2x\sqrt{2x+2h}+\sqrt{2x}(2x+2h)}\right) \times \frac{1}{h}\right]$$

$$= \lim_{h\to 0} \left[\left(\frac{-2h}{2x\sqrt{2x+2h}+\sqrt{2x}(2x+2h)}\right) \times \frac{1}{h}\right]$$

$$= \lim_{h\to 0} \left(\frac{-2}{2x\sqrt{2x+2h}+\sqrt{2x}(2x+2h)}\right)$$

$$= \frac{-2}{2x\sqrt{2x+2(0)}+\sqrt{2x}(2x+2(0))}$$

$$= \frac{-2}{2x\sqrt{2x}+\sqrt{2x}(2x)}$$

$$= \frac{-2}{4x\sqrt{2x}}$$

$$= \frac{-1}{2x\sqrt{2x}}$$

$$\therefore f'(x) = -\frac{1}{2x\sqrt{2x}}$$

(c) Find the x-coordinates of the maximum and minimum points of the curve

$$f(x) = 4x^3 + 7x^2 - 6x. \qquad [7]$$

$f(x) = 4x^3 + 7x^2 - 6x$

$f'(x) = 12x^2 + 14x - 6$

At stationary points, $f'(x) = 0$.

When $f'(x) = 0$,

$12x^2 + 14x - 6 = 0$

$(\div 2)$

$6x^2 + 7x - 3 = 0$

$(3x - 1)(2x + 3) = 0$

Either $3x - 1 = 0$ or $2x + 3 = 0$

$3x = 1$ $\qquad$ $2x = -3$

$x = \frac{1}{3}$ $\qquad$ $x = -\frac{3}{2}$

The x-coordinates of the turning points of the curve are $x = \frac{1}{3}$ and $x = -\frac{3}{2}$.

Now,

$f''(x) = 24x + 14$

$f'(x) = 12x^2 + 14x - 6$

When $x = \frac{1}{3}$,

$$f''\left(\frac{1}{3}\right) = 24\left(\frac{1}{3}\right) + 14$$

$$= 8 + 14$$

$$= 22 \quad (> 0)$$

Since $f''\left(\frac{1}{3}\right) > 0$, then there is a minimum at $x = \frac{1}{3}$.

When $x = -\frac{3}{2}$,

$$f''\left(-\frac{3}{2}\right) = 24\left(-\frac{3}{2}\right) + 14$$

$$= -36 + 14$$

$$= -22 \quad (< 0)$$

Since $f''\left(-\frac{3}{2}\right) < 0$, then there is a maximum at $x = -\frac{3}{2}$.

(d) A water tank is made by rotating the curve with equation $\frac{x^2}{4}+\frac{y^2}{25}=1$ about the x-axis between $x=0$ and $x=2$.

Find the volume of water that the tank can hold. [3]

$$\frac{x^2}{4}+\frac{y^2}{25}=1$$

$$\frac{y^2}{25}=1-\frac{x^2}{4}$$

$$y^2=25\left(1-\frac{x^2}{4}\right)$$

$$y^2=25-\frac{25}{4}x^2$$

Volume of water that the tank can hold $=\pi\int_a^b y^2\,dx$

$$=\pi\int_0^2\left(25-\frac{25}{4}x^2\right)dx$$

$$=\pi\left[25x-\left(\frac{25}{4}\right)\left(\frac{x^3}{3}\right)\right]_0^2$$

$$=\pi\left[25x-\frac{25x^3}{12}\right]_0^2$$

$$=\pi\left\{\left[25(2)-\frac{25(2)^3}{12}\right]-\left[25(0)-\frac{25(0)^3}{12}\right]\right\}$$

$$=\pi\left(50-\frac{50}{3}-0\right)$$

$$=\pi\left(\frac{100}{3}\right)$$

$$=\frac{100\pi}{3}\text{ units}^3$$

Total: 20 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.